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### Dual purpose energy plant

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#### Abstract

A system for generating electricity with reduced or negative carbon emissions. The system includes a power plant section having an electricity generating unit having an input coupled to a hydrocarbon fuel supply and an energy exchange path. The system also includes a direct air capture (DAC) section having a CO.sub.2 adsorption device having a CO.sub.2 adsorbent material and a ventilator electrically coupled to the electricity generating unit, the ventilator directing air flow through the CO.sub.2 adsorption device in a carbon capture mode, wherein the CO.sub.2 adsorption device is coupled to and in energy communication with the energy exchange path for releasing adsorbed CO.sub.2 in a carbon release mode.

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|-------------------|-------------------------------------------------------------------------|
| <b>Inventors:</b> | <b>Henning; Carlos (Buenos Aires, AR), Madden; David (Limerick, IE)</b> |
| <b>Applicant:</b> | <b>NUOVO PIGNONE TECNOLOGIE S.r.l. (Florence, IT)</b>                   |
| <b>Family ID:</b> | <b>1000008818520</b>                                                    |
| <b>Assignee:</b>  | <b>NUOVO PIGNONE TECNOLOGIE S.R.L. (Florence, IT)</b>                   |
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**References Cited**

**U.S. PATENT DOCUMENTS**

| Patent No.   | Issued Date | Patentee Name      | U.S. Cl.  | CPC          |
|--------------|-------------|--------------------|-----------|--------------|
| 9222480      | 12/2014     | Younes             | N/A       | F01N 5/02    |
| 9249690      | 12/2015     | Karni              | N/A       | F02C 3/34    |
| 10018115     | 12/2017     | Allam              | N/A       | F25J 3/04018 |
| 11168888     | 12/2020     | Boyce              | N/A       | F01D 15/10   |
| 11685658     | 12/2022     | GilroySmith        | 62/617    | F01K 13/00   |
| 11852376     | 12/2022     | Van der Walt       | N/A       | F24F 1/0059  |
| 12064725     | 12/2023     | Jang               | N/A       | B01D 53/185  |
| 2002/0023423 | 12/2001     | Viteri             | 60/39.182 | F02C 6/18    |
| 2013/0145773 | 12/2012     | Kulkarni           | 60/39.5   | F02C 3/34    |
| 2018/0340454 | 12/2017     | Smogorzewski       | N/A       | F02C 1/08    |
| 2021/0300765 | 12/2020     | Gilroysmith et al. | N/A       | N/A          |
| 2021/0381391 | 12/2020     | Ono                | N/A       | F02C 6/10    |
| 2021/0396176 | 12/2020     | Iwai               | N/A       | F01K 25/103  |
| 2022/0040668 | 12/2021     | Weston             | N/A       | N/A          |
| 2022/0401873 | 12/2021     | Eddaoudi           | N/A       | B01D 53/0462 |
| 2023/0173428 | 12/2022     | Sikka              | 95/113    | F03D 9/22    |
| 2023/0347276 | 12/2022     | Hillel             | N/A       | B01D 53/0462 |
| 2024/0058742 | 12/2023     | Dunn               | N/A       | B01D 53/261  |
| 2024/0075452 | 12/2023     | Lanigan-Atkins     | N/A       | B01J 20/3483 |
| 2024/0200492 | 12/2023     | Ferner             | N/A       | F02C 3/34    |

**FOREIGN PATENT DOCUMENTS**

| Patent No. | Application Date | Country | CPC |
|------------|------------------|---------|-----|
| 114282828  | 12/2021          | CN      | N/A |
| 114673571  | 12/2021          | CN      | N/A |
| 217410255  | 12/2021          | CN      | N/A |
| 2022235664 | 12/2021          | WO      | N/A |

**OTHER PUBLICATIONS**

Fahr et al., “Assessing the physical potential capacity of direct air capture with integrated supply of low-carbon energy sources”; Georgia Institute of Technology; Dec. 8, 2021; 26 pages. cited by applicant

Sadiq et al., “A Pilot-Scale Demonstration of Mobile Direct Air Capture Using Metal-Organic Frameworks”; Adv. Sustainable Syst. 2020, 4; 8 pages. cited by applicant

*Primary Examiner:* Bogue; Jesse S

*Attorney, Agent or Firm:* CANTOR COLBURN LLP

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## **Background/Summary**

### **BACKGROUND**

(1) This disclosure relates generally to a dual-purpose energy plant and more particularly to an energy plant that generates electricity with reduced or even negative carbon emissions as a whole.

(2) Electricity for an electric grid generally has several power plants that combust fossil fuel to produce energy for powering an electric generator connected to the electric grid. As is generally known, the combustion process emits carbon as a byproduct of the combustion. Unfortunately, the emitted carbon can contribute to atmospheric “greenhouse” gasses that can be a contributor to so called climate change. Climate change can be potentially dangerous as causing natural disasters or economic problems, which can result from a rise in sea level or agricultural problems for example. Hence, the power industry would welcome improvements in technology to generate electricity with reduced carbon emissions compared to conventional power plants or even negative carbon emissions where the electricity generation process as a whole removes more carbon from the atmosphere than any that may be added by combustion.

### **BRIEF SUMMARY**

(3) Disclosed is a system for generating electricity with reduced or negative carbon emissions. The system includes: a power plant section having an electricity generating unit having an input coupled to a hydrocarbon fuel supply and an energy exchange path; and a direct air capture (DAC) section having a CO.sub.2 adsorption device having a CO.sub.2 adsorbent material and a ventilator electrically coupled to the electricity generating unit, the ventilator directing air flow through the CO.sub.2 adsorption device in a carbon capture mode; wherein the CO.sub.2 adsorption device is coupled to and in energy communication with the energy exchange path for releasing adsorbed CO.sub.2 in a carbon release mode.

(4) Also disclosed is a method for generating electricity with reduced or negative carbon emissions. The method includes: generating the electricity in a power plant section of an electric power plant system using an electricity generating unit being coupled to a hydrocarbon fuel supply and having an electrical output and an energy exchange path; gathering by adsorption carbon dioxide (CO.sub.2) from air in a Direct Air Capture (DAC) section of the electric power plant system using a CO.sub.2 adsorption device having a CO.sub.2 adsorbent material in a carbon capture mode; flowing the air through the CO.sub.2 adsorption device using a ventilator receiving electric power from the electricity generating unit in the carbon capture mode; and releasing CO.sub.2 from the CO.sub.2 adsorption device by providing energy to the CO.sub.2 adsorbent material with energy from the energy exchange path in a carbon release mode.

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## **Description**

### **BRIEF DESCRIPTION OF THE DRAWINGS**

(1) The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:

- (2) FIG. 1 depicts aspects of a power plant section coupled to a direct air capture (DAC) section in a carbon capture mode;
- (3) FIG. 2 depicts aspect of a carbon dioxide adsorption device in the DAC section;
- (4) FIG. 3 depicts aspects of the power plant section coupled to the DAC section in a carbon release mode;
- (5) FIG. 4 depicts aspects of the power plant section coupled to the DAC section in the carbon release mode with the power plant section having an energy exchanger;
- (6) FIG. 5 depicts aspects of the power plant section having a supercritical CO<sub>2</sub> power cycle coupled to the DAC section in the carbon capture mode;
- (7) FIG. 6 depicts aspects of the power plant section having the supercritical CO<sub>2</sub> power cycle coupled to the DAC section in the carbon release mode;
- (8) FIG. 7 depicts aspects of the power plant section having the gas turbine and the heat recovery components coupled to the DAC section in the carbon capture mode;
- (9) FIG. 8 depicts aspects of the power plant section having the gas turbine and the heat recovery components coupled to the DAC section in the carbon release mode;
- (10) FIG. 9 depicts aspects of the power plant section having the gas turbine without the heat recovery components coupled to the DAC section in the carbon capture mode;
- (11) FIG. 10 depicts aspects of the power plant section having the gas turbine without the heat recovery components coupled to the DAC section in the carbon release mode;
- (12) FIG. 11 is a flowchart representation of a method for generating electricity with reduced or negative carbon emissions; and
- (13) FIG. 12 depicts aspect of a solid-oxide-fuel-cell (SOFC) system.
- (14) FIG. 13 depicts aspects of the SOFC system with enhancements in the carbon capture mode;
- (15) FIG. 14 depicts aspects of the SOFC system with enhancements in the carbon release mode; and
- (16) FIG. 15 is a flowchart representation of a method for generating electricity with reduced or negative carbon emissions using a fuel cell system.

#### DETAILED DESCRIPTION

- (17) A detailed description of one or more embodiments of the disclosed apparatus and method presented herein by way of exemplification and not limitation with reference to the figures, in which like elements are numbered alike.
- (18) Figures are presented below that illustrate various embodiments for generating electricity with reduced or negative carbon emissions. These figures include arrows for illustrating direction of fluid flow from a first component to a second component that is coupled to the first component. These arrows represent conduits, pipes, tubing, ducts or other types of flow paths for containing and directing the fluid flow. These arrows may also represent valves or dampers in the flow paths for controlling fluid flow in the flow paths in accordance with a mode of operation. These valves may be remotely controlled by a controller that may provide for automatic or manual operation. These arrows may also represent any pumps required for motivating fluid flow in accordance with a design configuration of the disclosed components. Arrows identifying electrical communication represent electrical conductors, transformers, switchgear, or other components needed for electrically powering a device. Locations where arrows leave or enter a component can represent output ports or input ports, respectively, for fluid flow or connections for electrical components. As can be seen in the figures, the arrows also indicate how a component is coupled, either directly or indirectly (with an intermediate component), to another component.
- (19) Disclosed herein are embodiments of an electric power plant system that generates electric power with reduced or negative carbon emissions. Hence, the electric power plant system has a dual purpose-generating electric power and reducing carbon emissions into the ambient air or even taking carbon out of the ambient air in total.
- (20) FIG. 1 illustrates a simplified diagram of an electric power plant system 10 having a power

plant section **11** coupled to a direct air capture (DAC) section **12** operating in a carbon capture mode. The power plant section **12** includes an electricity generating unit **9** for converting a fossil fuel or hydrocarbon fuel into electricity or electrical power. Non-limiting embodiments of the electricity generating unit **9** include a supercritical carbon dioxide power cycle, a gas turbine, a reciprocating engine (e.g., a gasoline or diesel engine), or a fuel cell system that produces a carbon dioxide stream and an energy discharge stream (e.g., a solid oxide fuel cell system). Each embodiment of the electricity generating unit **9** includes an energy exchange path (e.g., a heat discharge path) that may also be referred to as an engine exhaust in some embodiments. In one or more embodiments, the electricity generating unit **9** includes a prime mover or engine **13** that combusts the hydrocarbon fuel to provide mechanical energy to drive an electric generator **14** mechanically connected (such as by the illustrated shaft) or hydraulically connected to the engine **13**. The electric generator **14** generates electricity and a first portion of the generated electricity is provided to an electric grid by a grid connection. In embodiments using the fuel cell system, an inverter (not shown) connected to electrodes of the fuel cell system may be used to supply a first portion of alternating current (AC) electricity from the fuel cell system to the grid. A second portion of the generated electricity is provided to the DAC section **12** for operation of DAC section components. A hot gas recirculation line **15** receives hot gas from the engine **13** and provides the hot gas to an energy exchanger **16**. The energy exchanger **16** provides heat energy from the hot gas to the engine **13** in a first option or provides heat energy to an ambient environment in a second option. The cooler gas from the energy exchanger **16** passes through a first water absorber or water separator **17** to remove water from the gas. Optionally, a first compressor **18** compresses the drier gas from the water separator **17** and provides the compressed gas back to the engine **13**.

(21) The DAC section **12** includes a ventilator **19** that provides or directs ambient air to a carbon dioxide (CO.sub.2) adsorption device **20**. The CO.sub.2 adsorption device **20** removes CO.sub.2 and thus the associated carbon from the ambient air. Air having less CO.sub.2 is then discharged from the CO.sub.2 adsorption device **20**. The ventilator **19** is electrically coupled to and receives electric power from the electricity generating unit **9**. Hereinafter, any discussion of the electricity being supplied by the electric generator **14** inherently includes electricity being supplied by the fuel cell system when used as the electricity generating unit **9**.

(22) A controller **25** may be disposed in the power plant section **11** and/or the DAC section **12** for controlling operations of the electric power plant system **10** related to operating apparatus disclosed herein to enable electric power production with reduced or negative carbon emissions in accordance with the different modes of operation discussed herein. For example, the controller **25** may be configured to control valves or dampers for controlling a flow of fluid such as a working fluid, a heat transfer fluid, CO.sub.2 for sequestration or export or for recycling, or air. Also, the controller **25** may be configured to control electrical switchgear for controlling power to the ventilator **19** and/or CO.sub.2 compressors. The controller **25** may be configured to accept manual inputs and/or to provide automatic control. Automatic control may be implemented by an analog or digital processor implementing an algorithm. The algorithm may include model-based learning, machine learning, and/or artificial intelligence. In one or more embodiments, the algorithm may implement a neural network. The controller **25** may also include traditional control systems such as proportional, integral, and/or derivative (PID) control. Additionally, the controller **25** may receive inputs from sensors such as temperature, pressure, and/or flow sensors distributed throughout the electric power plant system **10**. Further, the controller **25** may be configured to communicate with other processing devices whether locally such as by wireless communication or remotely such as by over the internet. Accordingly, the controller **25** can be used to optimize the reduction of carbon emissions.

(23) FIG. 2 depicts aspects of the CO.sub.2 adsorption device **20**. In the embodiment of FIG. 2, the CO.sub.2 adsorption device **20** includes a housing **22** that supports a CO.sub.2 adsorbent material **21** and directs air flow from the ventilator **19** to the adsorbent material **21**. Heating the CO.sub.2

adsorbent material **21** in a carbon release mode causes the CO.sub.2 adsorbent material **21** to release CO.sub.2 for further processing and sequestration. In a non-limiting embodiment, the CO.sub.2 adsorbent material **21** is a metal-organic-framework (MOF) as known in the art.

(24) FIG. 3 illustrates a simplified diagram of the electric power plant system **10** having the power plant section **11** coupled to the DAC section **12** operating in a carbon release mode. In the carbon release mode, heat from the energy exchanger **16** is used to heat the CO.sub.2 adsorbent material **21** to release the adsorbed CO.sub.2. The heat can be provided indirectly with the energy exchanger **16** being a heat exchanger or directly by mass flow of the hot exhaust or emission emitted by the electricity generating unit **9** being directed directly to the CO.sub.2 adsorption device **20**. With the energy exchanger **16** being a heat exchanger, heat from a hot exhaust fluid on the primary side heats a heat transfer fluid on the secondary side, which is then used to heat the CO.sub.2 adsorbent material **21**. With the energy exchanger **16** providing heat directly to the CO.sub.2 adsorbent material **21**, the energy exchanger **16** includes components such as piping and valves to directly flow the hot exhaust fluid to the CO.sub.2 adsorption device **20** in a mode referred to as “mass flow.”

(25) CO.sub.2 and water discharged from the CO.sub.2 adsorption device **20** goes to a second water absorber or water separator **30** where water is separated from the entering CO.sub.2 and water. Dry CO.sub.2 from the second water separator **30** goes to the first compressor **18**. A first portion of compressed CO.sub.2 from the first compressor **18** is recycled back to the engine **13** while a second portion of the compressed CO.sub.2 is exported for sequestration. Optionally, a carbon export heat exchanger (HX) **31** may be coupled to an output of the first compressor **18** (e.g., disposed in a pipeline conveying the second portion of the compressed CO.sub.2 for export) to extract energy from the second portion of the compressed CO.sub.2 for heating the CO.sub.2 adsorbent material **21** in the carbon release mode. The second portion of the compressed CO.sub.2 flows through a primary side of the carbon export heat exchanger **31** to heat a heat transfer fluid in the secondary side of the carbon export heat exchanger **31**. This heat transfer fluid conveys energy to the CO.sub.2 adsorbent material **21** in the carbon release mode to aid in releasing the adsorbed CO.sub.2.

(26) FIG. 4 illustrates a simplified diagram of the electric power plant system **10** having the power plant section **11** coupled to the DAC section **12** operating in a carbon release mode (also referred to as regeneration). Here, heat is provided to the CO.sub.2 adsorber **20** to release adsorbed CO.sub.2 by a heat exchanger **40** disposed in the DAC section **12**. A primary side of the heat exchanger **40** is coupled to and receives heat energy from the energy exchanger **16**. A secondary side of the heat exchanger **40** is coupled to and provides heat to the CO.sub.2 adsorption device **20**. CO.sub.2 from the second water absorber or water separator **30** is compressed for export and sequestration by a second compressor **41** disposed in the DAC section **12**. The second compressor **41** is electrically coupled to and powered by the electric generator **14**.

(27) FIG. 5 illustrates a simplified diagram of the electric power plant system **10** having the power plant section **11** coupled to the DAC section **12** operating in a carbon capture mode with the engine **13** being a supercritical CO.sub.2 power cycle. In the embodiment of FIG. 5, the supercritical CO.sub.2 power cycle includes a combustor **50** configured to combust a hydrocarbon fuel to produce CO.sub.2 and water in a supercritical state. In one or more embodiments, the combustor **50** includes a combustion chamber (not shown) where fuel and oxidant are combined and the combustion takes place. The supercritical CO.sub.2 and water are provided to a turbo-expander **51**. A discharge of the turbo-expander **51** is coupled to the energy exchanger **16**, which provides a heat sink causing the supercritical CO.sub.2 and water to expand in the turbo-expander **51** to turn an output shaft that is either directly or indirectly coupled to the electric generator **14**. In general, the turbo-expander **51** includes a turbine having blades (not shown) upon which expanding fluid impinges to turn the turbine and a turbine shaft coupled to the electric generator **14**. An air separation unit (ASU) **52** is disposed in the power plant section **11** with an output coupled to the

combustor **50**. The ASU **52** is configured to separate oxygen (O.sub.2) from ambient air to provide the separated oxygen to the combustor **50** to aid in the combustion process.

(28) FIG. **6** illustrates a simplified diagram of the electric power plant system **10** having the power plant section **11** coupled to the DAC section **12** operating in a carbon release mode with the engine **13** being the supercritical CO.sub.2 power cycle. In the carbon release mode, heat is provided to the CO.sub.2 adsorption device **20** by a heat transfer fluid in a line or pipe to the energy exchanger **16**. The heat can be provided directly by mass flow or indirectly by the energy exchanger **16** being a heat exchanger where the secondary side provides the heat. In the embodiment of FIG. **6**, the second compressor **41** provides compressed CO.sub.2 to either an output line of the energy exchanger **16** or a separate input to the water separator **17**.

(29) FIG. **7** illustrates a more detailed embodiment of the electric power plant system **10** having the power plant section **11** coupled to the DAC section **12** operating in the carbon capture mode with the engine **13** being a gas turbine **90** (may also be a reciprocating engine or a fuel cell inherent with electricity generating electrodes or any engine having a heat and CO.sub.2 exhaust stream). Components not shown in either the power plant section **11** or the DAC section **12** may be considered to be disposed adjacent to the power plant section **11** or alternatively in the power plant section **11**. In the embodiment of FIG. **9**, a heat recovery power cycle (HRPC) is disposed in an exhaust path of the gas turbine **90** between an exhaust heat exchanger **95** and the gas turbine **90**. The HRPC uses waste energy from the gas turbine exhaust to generate more electricity, thus, increasing efficiency of the electric power generation. The heat recovery power cycle includes a heat recovery steam generator (HRSG) **91**, a steam turbine **92**, a HRPC heat exchanger **93**, and a HRPC pump **94**. The HRSG **91** generates steam using heat from the exhaust path of the gas turbine **90**. The generated steam is received by the steam turbine **92** to convert the energy of the steam to mechanical energy to turn a shaft coupled to the electric generator **14** or alternatively to another electric generator also connected to the electric grid. The HRPC heat exchanger **93** provides a heat sink for the discharge of the steam turbine **92** by condensing the steam to provide the necessary pressure drop across the steam turbine **92**. The HRPC pump **94** pumps the condensate from the HRPC heat exchanger **93** back to the HRSG **91** to complete the heat recovery power cycle.

(30) Downstream of the HRPC, the exhaust heat exchanger **95** further cools the exhaust from the gas turbine **90**. The exhaust is cooled by air or other cooling source in the secondary side of the exhaust heat exchanger **95**. From the exhaust heat exchanger **95**, a first portion of the exhaust flows to a CO.sub.2 capture device **97**, which captures CO.sub.2 from the exhaust. In one or more embodiments, the CO.sub.2 capture device **97** implements a Chilled Ammonia Process (CAP) or a Compact Carbon Capture with rotating bed (3C), all known in the art. The captured CO.sub.2 is then compressed by a first captured CO.sub.2 compressor **98** and dried by a captured CO.sub.2 water separator **99**. The dried captured CO.sub.2 is then further compressed by a second captured CO.sub.2 compressor **89** and provided for export and sequestration. A second portion of the exhaust discharged by the exhaust heat exchanger **95** is used for exhaust gas recirculation (EGR) and flows to a water separator **96** to separate water from the second portion of the exhaust to provide dry EGR exhaust. The dry EGR exhaust is then combined with air entering the gas turbine **90** for combustion.

(31) FIG. **8** illustrates a more detailed embodiment of the electric power plant system **10** having the power plant section **11** coupled to the DAC section **12** operating in the carbon release mode with the engine **13** being the gas turbine **90**. In the carbon release mode, both the exhaust heat exchanger **95** and the HRPC heat exchanger **93** supply heat to the CO.sub.2 adsorption device **20** to release (also referred to as regeneration) CO.sub.2 and supply the released CO.sub.2 to the first captured CO.sub.2 compressor **98**. Hence, in the carbon release mode the first captured CO.sub.2 compressor **98** compresses CO.sub.2 provided by the CO.sub.2 capture device **97** and the CO.sub.2 adsorption device **20**. In general, in the carbon release mode the controller **25** will turn off the ventilator **19** and open flow control devices to supply energy (e.g., heat) to the CO.sub.2 adsorption

device **20**.

(32) FIG. **9** illustrates another more detailed embodiment of the electric power plant system **10** having the power plant section **11** coupled to the DAC section **12** operating in the carbon capture mode with the engine **13** being the gas turbine **90**. The embodiment of FIG. **9** is similar to the embodiment of FIG. **7**, but without the heat recovery power cycle disposed downstream of gas turbine exhaust. Here, the gas turbine exhaust goes directly to the exhaust heat exchanger **95** where the exhaust is mainly cooled before going to the CO<sub>2</sub> capture device **97**, the first captured CO<sub>2</sub> compressor **98**, the captured CO<sub>2</sub> water separator **99**, and the second captured CO<sub>2</sub> compressor **89** in that order.

(33) FIG. **10** illustrates another more detailed embodiment of the electric power plant system **10** having the power plant section **11** coupled to the DAC section **12** operating in the carbon release mode with the engine **13** being the gas turbine **90**. Here, in the carbon release mode heat for releasing CO<sub>2</sub> from the adsorption device **20** is provided only or mostly by a heat transfer fluid flowing through a secondary side of the exhaust heat exchanger **95**. The heat transfer fluid is heated by heat energy in the gas turbine exhaust.

(34) FIG. **11** is a flow chart for a method **130** for generating electricity with reduced or negative carbon emissions. Block **131** calls for generating the electricity in a power plant section of an electric power plant system using an electricity generating unit being coupled to a hydrocarbon fuel supply and having an electrical output and an energy exchange path. In one or more embodiments, the energy exchange path operates as an energy discharge path. In non-limiting embodiments, the electricity generating unit operates in a supercritical CO<sub>2</sub> power cycle, as a gas turbine, as a reciprocating engine, or as a fuel cell system having an energy discharge stream and a CO<sub>2</sub> discharge stream.

(35) Block **132** calls for gathering by adsorption carbon dioxide (CO<sub>2</sub>) from air in a Direct Air Capture (DAC) section of the electric power plant system using a CO<sub>2</sub> adsorption device having a CO<sub>2</sub> adsorbent material in a carbon capture mode.

(36) Block **133** calls for flowing the air through the CO<sub>2</sub> adsorption device using a ventilator receiving electric power from the electricity generating unit in the carbon capture mode.

(37) Block **134** calls for releasing CO<sub>2</sub> from the CO<sub>2</sub> adsorption device by providing energy to the CO<sub>2</sub> adsorbent material with energy from the energy exchange path in a carbon release mode.

(38) In embodiments where the electricity generating unit operates in a supercritical CO<sub>2</sub> power cycle, the method **130** may also include: (1) combusting a hydrocarbon fuel using a combustor disposed in the supercritical CO<sub>2</sub> power cycle; (2) converting energy released by the combusting to mechanical output energy using an expander disposed in the supercritical CO<sub>2</sub> power cycle and coupled to an output of the combustor; (3) providing oxidant to the combustor using an air separation unit (ASU) coupled to an input the combustor; and (4) extracting water from working fluid flow after expansion in the expander using a water separation unit coupled to an output of the expander; wherein the electric generator is coupled to a mechanical output of the expander; and wherein the energy exchange path is a working fluid discharge path of the expander.

(39) The method **130** may also include using an energy exchanger coupled to a working fluid discharge path of the expander for supplying energy indirectly in a form of heat or directly in a form of CO<sub>2</sub> mass flow to the CO<sub>2</sub> adsorption device in the carbon release mode to release CO<sub>2</sub> from the CO<sub>2</sub> adsorbent material, wherein the CO<sub>2</sub> adsorbent material comprises metal-organic-framework (MOF).

(40) The method **130** may also include introducing a CO<sub>2</sub> flow from the DAC section in the carbon release mode into the combustor by using a first CO<sub>2</sub> compressor disposed in the DAC section and receiving electric power from the electrical generator for compressing the CO<sub>2</sub> released from the CO<sub>2</sub> adsorption material in the carbon release mode.



(41) The method **130** may also include compressing CO.sub.2 discharged from the energy exchanger and from the output of the first CO.sub.2 compressor using a second CO.sub.2 compressor disposed in the power plant section, receiving electric power from the electric generator, and coupled to an output of the water separator and an output of the first CO.sub.2 compressor, wherein a first portion of a discharge from the second CO.sub.2 compressor is exported for the sequestration and a second portion of the discharge from the second CO.sub.2 compressor is recycled into the combustor.

(42) In embodiments where the electricity generating unit includes a gas turbine, reciprocating engine, or fuel cell system each having an energy exhaust, the method **130** may include: (1) capturing CO.sub.2 from the energy exhaust using a CO.sub.2 capture unit coupled to the energy exhaust path; and (2) compressing the CO.sub.2 captured by the CO.sub.2 capture unit for exporting the compressed CO.sub.2 using a CO.sub.2 compressor coupled to an outlet of the CO.sub.2 capture unit, the CO.sub.2 compressor receiving electric power from the electricity generating unit. The CO.sub.2 capture unit may implement a Chilled Ammonia Process (CAP) or a Compact Carbon Capture with rotating bed (3C). The CO.sub.2 adsorbent material may include a metal-organic-framework (MOF). The method **130** may also include: (3) recovering heat from the energy exhaust path for generating steam using a heat recovery steam generator (HRSG) disposed in the power plant section in the energy exhaust path between the engine and the CO.sub.2 capture unit; and (4) generating the electricity using a steam turbine coupled to an electric generator, the steam turbine receiving steam from the HRSG. The method **130** may further include: (5) compressing the CO.sub.2 captured by the CO.sub.2 capture unit using a first CO.sub.2 compressor coupled to an outlet of the CO.sub.2 capture unit, the first CO.sub.2 compressor receiving electric power from the electricity generating unit; extracting water from compressed CO.sub.2 from the first CO.sub.2 compressor using a water separator coupled to an output of the first CO.sub.2 compressor to provide dry compressed CO.sub.2; and (6) compressing the dry compressed CO.sub.2 using a second CO.sub.2 compressor coupled to an output of the water separator, the second CO.sub.2 compressor receiving electric power from the electricity generating unit.

(43) As disclosed herein, the term “fuel cell” relates to a type of fuel cell or fuel cell system that produces a carbon dioxide stream and an energy discharge stream. One example of this type of fuel cell is a solid-oxide-fuel-cell (SOFC) system referred to as a SOFC **140** illustrated in FIG. **12**. The SOFC **140** includes a fuel cell reactor **141** that produces an electrical output to supply electricity to the grid and the DAC section **12**. Fluid outputs of the fuel cell reactor **141** are coupled to a combustor **142** and a reformer **143**. Chemical reactions for the fuel cell reactor **141** and the reformer **143** are also illustrated in FIG. **12**. The combustor **142** combusts unutilized hydrocarbon fuel from the fuel cell reactor **141**. Output from the combustor **142** provides the energy discharge stream or energy exchange path. The reformer **143** performs a steam reforming reaction using heat and steam from the fuel cell reactor **141** to reform a hydrocarbon fuel to provide hydrogen to the fuel cell reactor **141** where the hydrogen is oxidized at an electrode to produce electricity.

(44) FIG. **13** illustrates a simplified diagram of the electric power plant system **10** having the power plant section **11** coupled to the DAC section **12** operating in a carbon capture mode with the electricity generating unit **9** being the SOFC **140** integrated with enhancements that include an expander. In this embodiment, the electricity generating unit **9** includes the SOFC system **140**, an electric motor (EM) **150**, a low pressure (LP) air compressor **151**, a high pressure (HP) compressor **152**, an expander **153**, and a multi-stream heat exchanger (HX) **154**. The EM **150** is coupled to the LP compressor **151** for driving the LP compressor **151** and receives electric power from the SOFC system **140**. During start-up a local battery may be used to temporarily power the EM **150** until the fuel cell reactor begins to generate the electricity. Compressed air from the LP compressor **151** is provided to an input of the HP compressor **152**. The terms LP and HP may be considered as relative terms such that the pressure output of the HP compressor **152** is higher than the pressure output of the LP compressor **151**. The expander **153** is coupled to the HP compressor **152** for driving the HP

compressor **152**. The multi-stream heat exchanger (HX) **154** has a primary side and at least two secondary sides heated by the primary side. A primary side input is coupled to a high pressure and high temperature exhaust stream of the SOFC system **140**, while a primary side output is coupled to an input of the expander **153**. Hence, the high pressure and high temperature exhaust stream of the SOFC system **140** powers the expander **153** and thus the HP compressor **152**. One secondary side of the HX **154** is coupled to a hydrocarbon fuel input to the SOFC system **140** for heating the hydrocarbon fuel to the SOFC **140**. Another secondary side of the HX **154** is coupled to a HP output of the HP compressor **152** for heating an oxidant supply to the SOFC system **140**.

(45) The SOFC system **140** integrated with the above components are configured to produce electric power that is provided to the grid, to the EM **150**, and to the DAC components. The SOFC system **140** produces electrical power when a fuel and oxidant undergo an electrochemical reaction inside the SOFC **140** under certain defined conditions. The expander produces mechanical power by converting work of high pressure and temperature exhaust gas of the SOFC system **140** that is expanded to low temperature and low pressure. The LP and HP compressors **151** and **152** are configured to deliver high pressure oxidant (e.g., air) to the SOFC system **140**. The SOFC system **140** is configured to receive compatible hydrocarbon fuels such as natural gas for example. The heat exchanger **154** is external to the SOFC system **140** and is configured to allow heat exchange between 3 streams—(1) a fuel stream of a SOFC fuel feed line, upstream the SOFC **140**, (2) the exhaust gas from the SOFC **140**, downstream of SOFC **140**, and (3) a high-pressure oxidant feed line to the SOFC **140**, upstream of SOFC **140**.

(46) The electrical power is generated by the pressurized fuel cell reactor **141** performing an electrochemical reaction between pre-heated fuel and oxidant streams coming from the multi-stream heat exchanger **154**. A combustion chamber of the combustor **142** is an integral part of SOFC **140**, that helps in burning unutilized fuel from the pressurized fuel cell reactor **141** by use of oxidant, producing high temperature exhaust gases. The hot exhaust gases from the fuel cell reactor **141** exchange heat with fuel and oxidant streams in the multi-stream heat exchanger **154**. The expander **153** is arranged down stream of heat exchanger **154** and is driven by hot exhaust gases of SOFC system **140**. The expander **153** and HP compressor **152** are connected to each other through a common shaft in one or more embodiments. The power produced by the expander **153** is used to drive the HP compressor **152** and excess power may be extracted through a generator (not shown) connected to the other side of the shaft of the expander **153**.

(47) The power producing unit, the solid oxide fuel cell system **140** integrated with the expander **153** operates as follows. Air to the SOFC system **140** is compressed through a two-stage compression system. An air stream is drawn from atmosphere by the low-pressure compressor **151**, which is driven by the small electrical motor **150**, is compressed to a first stage of pressure. The air from the first stage of compression is fed to the high-pressure compressor **152**, which is driven by the gas expander **153**, and the air is further compressed to high pressure. A compressed air stream from the HP compressor **152** is passed through the heat exchanger **154** before being introduced in to SOFC system **140**. The heat exchanger **154** also heats a fuel stream, the source of power, from an outside source. The heated stream of air and the heated stream of fuel from the heat exchanger are injected in to the SOFC system **140**. Inside the fuel cell reactor **141**, the heated air and the heated fuel undergo an electro-chemical reaction at the electrodes of the fuel cell reactor **141** to produce electrical power. The exhaust gases, unutilized fuel and air are mixed in the integral combustor **142** of the SOFC **140** and combusted in the combustion chamber. This energy of combustion further increases the temperature of the exhaust gas stream from the combustor **142**. The high temperature exhaust gas stream is then fed to the heat exchanger **154** to exchange heat with incoming streams of fuel and air. From the heat exchanger **154**, the high pressure and high temperature exhaust gas stream is fed to the expander **153**. The high pressure and high temperature exhaust gas stream undergoes expansion in the expander **153** converting its energy to work and further optionally to electrical power. The expander **153** transmits part power to HP compressor

152 and optionally remaining power to a generator through common shaft.

(48) FIG. 14 illustrates a simplified diagram of the electric power plant system 10 having the power plant section 11 coupled to the DAC section 12 operating in a carbon release mode with the electricity generating unit 9 being the SOFC system 140 integrated with the expander 153. In the carbon release mode, energy (i.e., heat) is directed from the heat exchanger 95 to the CO.sub.2 adsorption device 20 to heat the CO.sub.2 adsorption device 20. The CO.sub.2 released from the CO.sub.2 adsorption device 20 is directed to the first captured CO.sub.2 compressor 98 for further processing prior to being exported for sequestration.

(49) FIG. 15 is a flow chart for a method 170 for generating electricity with reduced or negative carbon emissions using a fuel cell system. Block 171 calls for generating the electricity in a power plant section using a solid oxide fuel cell (SOFC) system, the SOFC system: receiving hydrogen fuel and an air; reacting hydrogen and the air for generating the electricity; combusting unutilized fuel to provide an energy exchange path; reforming a hydrocarbon fuel from a hydrocarbon fuel supply using heat and steam from the fuel cell reactor to provide the hydrogen fuel to the SOFC reactor; and compressing air to provide the compressed air to the SOFC reactor. In one or more embodiments, the energy exchange path operates as an energy discharge path.

(50) Block 172 calls for gathering by adsorption carbon dioxide (CO.sub.2) from air in a direct air capture (DAC) section using a CO.sub.2 adsorption device having a CO.sub.2 adsorbent material in a carbon capture mode.

(51) Block 173 calls for flowing the air through the CO.sub.2 adsorption device using a ventilator receiving electric power from the SOFC system in the carbon capture mode.

(52) Block 174 calls for releasing CO.sub.2 from the CO.sub.2 adsorption device by providing energy to the CO.sub.2 adsorbent material from the energy exchange path in a carbon release mode. The releasing may include at least one of heating the CO.sub.2 adsorbent material using mass flow of a fluid in the energy exchange path or heating the CO.sub.2 adsorbent material using a heat transfer fluid heated by a heat exchanger in a secondary side, a primary side of the heat exchanger being heated by energy from the energy exchange path.

(53) The method 170 may also include capturing CO.sub.2 using a CO.sub.2 capture device receiving an exhaust from the combustor.

(54) The method 170 may also include: compressing CO.sub.2 released by the CO.sub.2 adsorbent material and the CO.sub.2 capture device to provide compressed CO.sub.2; separating water from the compressed to provide dry compressed CO.sub.2; compressing the dry compressed CO.sub.2 to provide further compressed dry CO.sub.2; and exporting the further compressed dry CO.sub.2.

(55) The method 170 may also include: driving a high-pressure compressor providing the compressed air to the SOFC reactor using an expander coupled to the high-pressure compressor and driven by exhaust from the combustor; and reforming the hydrocarbon fuel using energy from the SOFC reactor.

(56) The method 170 may also include providing low-pressure compressed air to the high-pressure compressor using a low-pressure compressor; and driving the low-pressure compressor using an electric motor coupled to the low-pressure compressor.

(57) The method 170 may also include: drying exhaust from the expander to provide dried exhaust; and recycling the dried exhaust into the low-pressure compressor.

(58) In support of the teachings herein, various analysis components may be used, including a digital and/or an analog system. For example, the controller 25 may include digital and/or analog systems. The system may have components such as a processor, storage media, memory, input, output, communications link (wired, wireless, optical or other), user interfaces (e.g., a display or printer), software programs, signal processors (digital or analog) and other such components (such as resistors, capacitors, inductors and others) to provide for operation and analyses of the apparatus and methods disclosed herein in any of several manners well-appreciated in the art. It is considered that these teachings may be, but need not be, implemented in conjunction with a set of computer

executable instructions stored on a non-transitory computer readable medium, including memory (ROMs, RAMs), optical (CD-ROMs), or magnetic (disks, hard drives), or any other type that when executed causes a computer to implement the method of the present invention. These instructions may provide for equipment operation, control, data collection and analysis and other functions deemed relevant by a system designer, owner, user or other such personnel, in addition to the functions described in this disclosure.

(59) Batteries to supply electric power for start-up or shutdown purposes may be disposed in various locations throughout the electric power plant system **10**. The batteries may be charged using electricity generated in the power plant section **11**.

(60) Set forth below are some embodiments of the foregoing disclosure:

(61) Embodiment 1. A system for generating electricity with reduced or negative carbon emissions, including a power plant section comprising an electricity generating unit having an input coupled to a hydrocarbon fuel supply and an energy exchange path, a direct air capture (DAC) section comprising a CO.sub.2 adsorption device having a CO.sub.2 adsorbent material and a ventilator electrically coupled to the electricity generating unit, the ventilator directing air flow through the CO.sub.2 adsorption device in a carbon capture mode, wherein the CO.sub.2 adsorption device is coupled to and in energy communication with the energy exchange path for releasing adsorbed CO.sub.2 in a carbon release mode.

(62) Embodiment 2. The system as in any prior embodiment, further including a CO.sub.2 compressor coupled to a CO.sub.2 release port of the CO.sub.2 adsorption device for compressing a portion of the adsorbed CO.sub.2 in a carbon release mode and providing the portion for export, and a carbon export heat exchanger having a primary side coupled to an output of the CO.sub.2 compressor and a secondary side coupled to the CO.sub.2 adsorption device for heating the CO.sub.2 adsorbent material in the carbon release mode.

(63) Embodiment 3. The system as in any prior embodiment, wherein the electricity generating unit comprises components of a supercritical CO.sub.2 power cycle, a gas turbine, a reciprocating engine, or a fuel cell system producing a CO.sub.2 stream and an energy discharge stream.

(64) Embodiment 4. The system as in any prior embodiment, wherein the components of the supercritical CO.sub.2 power cycle comprise a combustor disposed in the thermodynamic power cycle to combust a hydrocarbon fuel, an air separation unit (ASU) coupled to an input the combustor for providing oxidant to the combustor, an expander coupled to an output of the combustor for converting energy released by the combusting to mechanical output energy and comprising a working fluid discharge path as the energy exchange path, an electric generator coupled to the expander to receive the mechanical output energy from the expander, and a water separation unit coupled to the working fluid discharge path of the expander to extract water from working fluid flow after expansion of the working fluid in the expander.

(65) Embodiment 5. The system as in any prior embodiment, wherein the components further comprise an energy exchanger coupled to the working fluid discharge path of the expander to supply energy indirectly in a form of heat from a heat exchanger or directly in a form CO.sub.2 mass flow to the CO.sub.2 absorption device in the carbon release mode to release CO.sub.2 from the CO.sub.2 absorbent material.

(66) Embodiment 6. The system as in any prior embodiment, further including a first CO.sub.2 compressor disposed in the DAC section, electrically coupled to the electric generator, having an input in fluid communication with the CO.sub.2 released from the CO.sub.2 adsorbent material in the carbon release mode to compress the released CO.sub.2 to provide compressed CO.sub.2, and having an output coupled to the supercritical CO.sub.2 power cycle to introduce the compressed CO.sub.2 into the supercritical CO.sub.2 power cycle.

(67) Embodiment 7. The system as in any prior embodiment, further including a second CO.sub.2 compressor disposed in the power plant section, electrically coupled to the electric generator, having an input coupled to an output of the water separator and an output of the first CO.sub.2

compressor to compress CO.sub.2 discharged from the energy exchanger and from the first CO.sub.2 compressor, and having an output coupled to a carbon sequestration process and to the supercritical CO.sub.2 power cycle to recycle a portion of the CO.sub.2.

(68) Embodiment 8. The system as in any prior embodiment, wherein the CO.sub.2 adsorbent material comprises a metal-organic-framework (MOF).

(69) Embodiment 9. The system as in any prior embodiment, wherein the electricity generating unit comprises the gas turbine, the reciprocating engine, or the fuel cell system, each having a combustion exhaust path as the energy exchange path, the system further including a CO.sub.2 capture unit coupled to the combustion exhaust path for capturing CO.sub.2 from the combustion exhaust path, and a CO.sub.2 compressor and coupled to an outlet of the CO.sub.2 capture unit for compressing the CO.sub.2 captured by the CO.sub.2 capture unit.

(70) Embodiment 10. The system as in any prior embodiment wherein the CO.sub.2 capture unit comprises a chilled ammonia process (CAP) or a compact carbon capture with rotating bed (3C) and the CO.sub.2 adsorbent material comprises a metal-organic-framework (MOF).

(71) Embodiment 11. The system as in any prior embodiment, further including components of a heat recovery power cycle, the components including a heat recovery steam generator (HRSG) disposed in the power plant section in the combustion exhaust path downstream of an exhaust port of the electricity generating unit and upstream of the CO.sub.2 capture unit, the HRSG generating steam from heat of the engine exhaust path, a steam turbine coupled to the HRSG for receiving the generated steam and having a mechanical output coupled to the electric generator or another electric generator for generating the electricity, and an energy recovery energy exchanger disposed in the heat recovery power cycle downstream of the steam turbine and in energy communication with the CO.sub.2 adsorption device during the carbon release mode.

(72) Embodiment 12. The system as in any prior embodiment, wherein the fuel cell system includes a fuel cell reactor that performs an electro-chemical reaction of hydrogen and air to generate the electricity, the fuel cell reactor having an input that receives the hydrogen and air, an exhaust output that discharges unutilized hydrogen, a steam exhaust output that discharges steam, and a heat exhaust output that discharges energy, a combustor coupled to the exhaust output and the steam exhaust output, the combustor combusting the unutilized hydrogen, a combustion exhaust path of the combustor being the energy exchange path, and a reformer coupled to the heat exhaust output and the steam exhaust output of the fuel cell reactor, the reformer using the energy and the steam to reform a hydrocarbon fuel to provide the hydrogen.

(73) Embodiment 13. A method for generating electricity with reduced or negative carbon emissions, the method including generating the electricity in a power plant section of an electric power plant system using an electricity generating unit being coupled to a hydrocarbon fuel supply and having an electrical output and an energy exchange path, gathering by adsorption carbon dioxide (CO.sub.2) from air in a Direct Air Capture (DAC) section of the electric power plant system using a CO.sub.2 adsorption device having a CO.sub.2 adsorbent material in a carbon capture mode, flowing the air through the CO.sub.2 adsorption device using a ventilator receiving electric power from the electricity generating unit in the carbon capture mode, and releasing CO.sub.2 from the CO.sub.2 adsorption device by providing energy to the CO.sub.2 adsorbent material with energy from the energy exchange path in a carbon release mode.

(74) Embodiment 14. The method as in any prior embodiment, further including compressing the CO.sub.2 released from the CO.sub.2 adsorption device using a CO.sub.2 compressor to provide compressed CO.sub.2 for export, extracting heat from the CO.sub.2 for export using a carbon export heat exchanger coupled to the CO.sub.2 compressor to provide extracted heat, and heating the CO.sub.2 adsorbent material with the extracted heat to release CO.sub.2 from the CO.sub.2 adsorption device in the carbon release mode.

(75) Embodiment 15. The method as in any prior embodiment, wherein the electricity generating unit operates in a supercritical CO.sub.2 power cycle, as a gas turbine, as a reciprocating engine, or

a fuel cell system producing a carbon dioxide stream and an energy discharge stream.

(76) Embodiment 16. The method as in any prior embodiment, wherein the electricity generating unit operates in the supercritical CO<sub>2</sub> power cycle and the method further includes combusting a hydrocarbon fuel using a combustor disposed in the supercritical CO<sub>2</sub> power cycle, converting energy released by the combusting to mechanical output energy using an expander disposed in the supercritical CO<sub>2</sub> power cycle and having an input coupled to an output of the combustor, the expander being coupled to an electric generator for generating the electricity, providing oxidant to the combustor using an air separation unit (ASU) coupled to an input the combustor, extracting water from working fluid flow after expansion in the expander using a water separation unit coupled to an output of the expander, wherein the energy exchange path is a working fluid discharge path of the expander.

(77) Embodiment 17. The method as in any prior embodiment, further including using an energy exchanger coupled to the working fluid discharge path of the expander for supplying energy indirectly in a form of heat or directly in a form of CO<sub>2</sub> mass flow to the CO<sub>2</sub> adsorption device in the carbon release mode to release CO<sub>2</sub> from the CO<sub>2</sub> adsorbent material, wherein the CO<sub>2</sub> adsorbent material comprises metal-organic-framework (MOF).

(78) Embodiment 18. The method as in any prior embodiment, further including introducing a CO<sub>2</sub> flow from the DAC section in the carbon release mode into the combustor by using a first CO<sub>2</sub> compressor disposed in the DAC section and receiving electrical power from the electric generator for compressing the CO<sub>2</sub> released from the CO<sub>2</sub> adsorption material in the carbon release mode.

(79) Embodiment 19. The method as in any prior embodiment, further including compressing CO<sub>2</sub> discharged from the energy exchanger and from the output of the first CO<sub>2</sub> compressor using a second CO<sub>2</sub> compressor disposed in the power plant section, receiving electric power from the electric generator, and coupled to an output of the water separator and an output of the first CO<sub>2</sub> compressor, wherein a first portion of a discharge from the second CO<sub>2</sub> compressor is exported for the sequestration and a second portion of the discharge from the second CO<sub>2</sub> compressor is recycled into the combustor.

(80) Embodiment 20. The method as in any prior embodiment, wherein the electricity generating unit operates as the gas turbine, the reciprocating engine, or the fuel cell system, each having a combustion exhaust path as the energy exchange path, the method further including capturing CO<sub>2</sub> from the combustion exhaust path using a CO<sub>2</sub> capture unit coupled to the combustion exhaust path, and compressing the CO<sub>2</sub> captured by the CO<sub>2</sub> capture unit for exporting the compressed CO<sub>2</sub> using a CO<sub>2</sub> compressor coupled to an outlet of the CO<sub>2</sub> capture unit, the CO<sub>2</sub> compressor receiving electric power from the electric generating unit.

(81) Embodiment 21. The method as in any prior embodiment, wherein the CO<sub>2</sub> capture unit comprises a Chilled Ammonia Process (CAP) or a Compact Carbon Capture with rotating bed (3C), and the CO<sub>2</sub> adsorbent material includes a metal-organic-framework (MOF).

(82) Embodiment 22. The method as in any prior embodiment, further including recovering energy from the combustion exhaust path for generating steam using a heat recovery steam generator (HRSG) disposed in the power plant section in the combustion exhaust path between the electricity generating unit and the CO<sub>2</sub> capture unit, and generating the electricity using a steam turbine coupled to the electric generator or another electric generator, the steam turbine receiving steam from the HRSG.

(83) Embodiment 23. The method as in any prior embodiment, wherein the compressing the CO<sub>2</sub> captured by the CO<sub>2</sub> capture unit includes compressing the CO<sub>2</sub> captured by the CO<sub>2</sub> capture unit using a first CO<sub>2</sub> compressor coupled to an outlet of the CO<sub>2</sub> capture unit, the first CO<sub>2</sub> compressor receiving electric power from the electricity generating unit, extracting water from compressed CO<sub>2</sub> from the first CO<sub>2</sub> compressor using a water

separator coupled to an output of the first CO.sub.2 compressor to provide dry compressed CO.sub.2, and compressing the dry compressed CO.sub.2 using a second CO.sub.2 compressor coupled to an output of the water separator, the second compressor receiving electric power from the electricity generating unit.

(84) Elements of the embodiments have been introduced with either the articles “a” or “an.” The articles are intended to mean that there are one or more of the elements. The terms “including” and “having” and the like are intended to be inclusive such that there may be additional elements other than the elements listed. The conjunction “or” when used with a list of at least two terms is intended to mean any term or combination of terms. The term “configured” relates to one or more structural limitations of a device that are required for the device to perform the function or operation for which the device is configured. The term “coupled” relates to being coupled directly or indirectly using an intermediate device. The terms “first” and “second” and the like are used to distinguish terms and not to denote a particular order.

(85) The flow diagram depicted herein is just an example. There may be many variations to this diagram or the steps (or operations) described therein without departing from the scope of the invention. For example, operations may be performed in another order or other operations may be performed at certain points without changing the specific disclosed sequence of operations with respect to each other. All of these variations are considered a part of the claimed invention.

(86) The disclosure illustratively disclosed herein may be practiced in the absence of any element which is not specifically disclosed herein.

(87) While one or more embodiments have been shown and described, modifications and substitutions may be made thereto without departing from the scope of the invention. Accordingly, it is to be understood that the present invention has been described by way of illustrations and not limitation.

(88) It will be recognized that the various components or technologies may provide certain necessary or beneficial functionality or features. Accordingly, these functions and features as may be needed in support of the appended claims and variations thereof, are recognized as being inherently included as a part of the teachings herein and a part of the invention disclosed.

(89) While the invention has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is intended that the invention not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the claims. Also, in the drawings and the description, there have been disclosed exemplary embodiments of the invention and, although specific terms may have been employed, they are unless otherwise stated used in a generic and descriptive sense only and not for purposes of limitation, the scope of the invention therefore not being so limited.

## Claims

1. A system for generating electricity provided to an electric grid with reduced or negative carbon emissions, the system comprising: a power plant section comprising an electricity generating unit comprising an energy exchange path, wherein the electricity generating unit comprises: a working fluid comprising carbon dioxide (CO.sub.2); a combustor coupled to a supply of hydrocarbon fuel and a supply of pure oxygen and configured to combust the hydrocarbon fuel with the pure oxygen to heat the CO.sub.2 to a supercritical state and to produce additional CO.sub.2; an expander coupled to an output of the combustor to receive the CO.sub.2 in the supercritical state and configured to convert energy of the CO.sub.2 in the supercritical state to mechanical output energy,

the expander comprising a working fluid discharge path as the energy exchange path; an electric generator coupled to the expander and configured to receive the mechanical output energy for generating the electricity, the electric generator comprising an output coupled to the electric grid; an energy exchanger comprising an input coupled to the expander and configured to receive CO.sub.2 discharged from the expander; a compressor comprising an input coupled to an output of the energy exchanger and an output coupled to an input of the combustor to compress and recycle at least a portion of the CO.sub.2 discharged by the expander; a direct air capture (DAC) section comprising a CO.sub.2 adsorption device having a CO.sub.2 adsorbent material and a ventilator electrically coupled to the electric generator, the ventilator directing air flow through the CO.sub.2 adsorption device in a carbon capture mode; wherein the CO.sub.2 adsorption device is coupled to and in energy communication with the energy exchange path for releasing adsorbed CO.sub.2 in a carbon release mode further comprising a water separator disposed in a line between the energy exchanger and the compressor and configured to separate water from a discharge of the energy exchanger; further comprising another compressor disposed in a line between the CO.sub.2 adsorption device and at least one of an input of the water separator or a line leading to an input of the water separator, the another compressor comprising an input coupled to an output of the CO.sub.2 adsorption device configured to discharge CO.sub.2 in the carbon release mode.

2. The system according to claim 1, further comprising: a CO.sub.2 compressor coupled to a CO.sub.2 release port of the CO.sub.2 adsorption device for compressing a portion of the adsorbed CO.sub.2 in a carbon release mode and providing the portion for export; and a carbon export heat exchanger having a primary side coupled to an output of the CO.sub.2 compressor and a secondary side coupled to the CO.sub.2 adsorption device for heating the CO.sub.2 adsorbent material in the carbon release mode.

3. The system according to claim 1, wherein the CO.sub.2 adsorbent material comprises a metal-organic-framework (MOF).

4. The system according to claim 1, wherein the CO.sub.2 adsorption device is coupled to the working fluid discharge path of the expander such that the CO.sub.2 adsorbent material is heated in the carbon release mode by CO.sub.2 discharged from the expander.

5. The system according to claim 1, further comprising a heat exchanger having a first side coupled to the working fluid discharge path of the expander and a second side coupled to the CO.sub.2 adsorption device such that the CO.sub.2 adsorbent material is heated in the carbon release mode by a heat transfer fluid heated in the heat exchanger by CO.sub.2 discharged from the expander.

6. A method for generating electricity provided to an electric grid with reduced or negative carbon emissions, the method comprising: using carbon dioxide (CO.sub.2) as a working fluid in an electricity generating unit of a power plant section of an electric power plant system; combusting hydrogen fuel and pure oxygen in a combustor coupled to a supply of hydrocarbon fuel and a supply of pure oxygen to heat the CO.sub.2 to a supercritical state and to produce additional CO.sub.2; converting energy of the CO.sub.2 in the supercritical state to mechanical output energy using an expander coupled to an output the combustor; generating the electricity using an electric generator coupled to the expander to receive the mechanical output energy and to provide the electricity to the electric grid; using a working fluid discharge path of the expander as an energy exchange path; receiving CO.sub.2 discharged from the expander using an energy exchanger comprising an input coupled to the expander to receive CO.sub.2 discharged from the expander; compressing and recycling at least a portion of the CO.sub.2 discharged by the expander to the combustor using a compressor comprising an input coupled to an output of the energy exchanger and an output coupled to an input of the combustor; gathering by adsorption carbon dioxide (CO.sub.2) from air in a Direct Air Capture (DAC) section of the electric power plant system using a CO.sub.2 adsorption device having a CO.sub.2 adsorbent material in a carbon capture mode; flowing the air through the CO.sub.2 adsorption device using a ventilator receiving electric power from the electric generator in the carbon capture mode; and releasing CO.sub.2 from the CO.sub.2



adsorption device by providing energy to the CO.sub.2 adsorbent material with energy from the energy exchange path in a carbon release mode further comprising separating water from a discharge comprising CO.sub.2 and water from the energy exchanger using a water separator having an output coupled to the input of the compressor; further comprising compressing CO.sub.2 output from the CO.sub.2 adsorption device in the carbon release mode using another compressor and providing compressed CO.sub.2 from the another compressor to at least one of an input of the water separator or a line leading to an input of the water separator.

7. The method according to claim 6, further comprising: compressing the CO.sub.2 released from the CO.sub.2 adsorption device using a CO.sub.2 compressor to provide compressed CO.sub.2 for export; extracting heat from the CO.sub.2 for export using a carbon export heat exchanger coupled to the CO.sub.2 compressor to provide extracted heat; and heating the CO.sub.2 adsorbent material with the extracted heat to release CO.sub.2 from the CO.sub.2 adsorption device in the carbon release mode.

8. The method according to claim 6, wherein releasing CO.sub.2 from the CO.sub.2 adsorption device by providing energy to the CO.sub.2 adsorbent material with energy from the energy exchange path in a carbon release mode comprises heating the CO.sub.2 adsorbent material with CO.sub.2 discharged from the expander.

9. The method according to claim 6, wherein releasing CO.sub.2 from the CO.sub.2 adsorption device by providing energy to the CO.sub.2 adsorbent material with energy from the energy exchange path in a carbon release mode comprises heating the CO.sub.2 adsorbent material with a heat transfer fluid heated in a second side of a heat exchanger that receives CO.sub.2 discharged from the expander in a first side of the heat exchanger.

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